

Toxic Category Classifier – Summary

Objective

Build an LSTM text classifier that assigns each input to one of 5 toxic-category labels: **Safe, Violent Crimes, Non-Violent Crimes, unsafe, Suicide & Self-Harm**. The pipeline covers data cleaning and label correction, leakage-safe preprocessing, and model training with class-imbalance handling.

Data Preparation

The source data started as 3,000 rows across two columns (image description, query). These were stacked into a single text column, producing 6,000 rows against a 9-class Toxic Category target.

Wrong labels. Cross-checking exact-duplicate texts against their assigned labels surfaced two kinds of errors:

- Duplicate rows with conflicting labels (the same text tagged with several different categories across the dataset).
- Near-duplicate template phrasings scattered inconsistently across categories.

Each case was corrected using a mix of LLM-assisted review and manual judgment, choosing the label best supported by the text's actual content and by how visibly-similar phrasings were labeled elsewhere in the data.

Duplicates. Exact duplicate rows were removed after label correction, so every row reflects a distinct text with a verified label.

Class imbalance after dedup. With duplicates removed, several of the original 9 classes had almost no unique examples left (as few as 1–3 rows), making them unusable for stratified splitting and training. Related classes were merged into broader, better-supported categories:

Original class	Merged into
Child Sexual Exploitation	Sex-Related Crimes
Elections	Non-Violent Crimes
Sex-Related Crimes	unsafe

This brought the target down from 9 classes to **5 final classes**: Safe, Violent Crimes, Non-Violent Crimes, unsafe, Suicide & Self-Harm. The resulting class distribution (unique rows) was:

Class	Rows
Safe	868
Violent Crimes	689
Non-Violent Crimes	234
unsafe	227
Suicide & Self-Harm	3

Suicide & Self-Harm remains extremely small; it was kept separate rather than merged further because it needs a different handling path (e.g. crisis response) than a generic harmful-content label, but its low count limits how reliably the model can learn or be evaluated on it.

Text Preprocessing

Cleaning. Basic text normalization was applied before any splitting: removing hashtags, email addresses, URLs, and collapsing extra whitespace, along with related noise-reduction steps (punctuation, casing).

Split. Data was split into train / validation / test sets before any further preprocessing, so that steps like tokenization and embedding lookup were fit only on the training set and applied to validation/test — avoiding data leakage.

Label encoding. The target Toxic Category was label-encoded into integer class ids for the 5 final classes.

Tokenization & padding. Text was tokenized into integer sequences (vocabulary built from the training set only) and padded/truncated to a fixed max length so all inputs have equal shape for the LSTM.

Embedding. Token ids are mapped to dense vectors via an embedding layer, giving the LSTM a learned (or pretrained) representation of each word rather than raw one-hot/token ids.

Model Architecture & Training Setup

An LSTM/RNN-based architecture was used on top of the embedding layer, with the following training configuration:

Component	Choice
Core layers	LSTM / RNN layers over the embedded sequence
Optimizer	AdamW, with weight decay for regularization

LR schedule	ReduceLROnPlateau (lowers the learning rate when validation performance stalls)
Early stopping	Training halted once validation performance stopped improving, restoring best weights
Class imbalance	Class weights applied during training to compensate for the uneven class sizes (see Data Preparation)

This combination targets the two main risks in the dataset: overfitting on a relatively small number of unique examples, and the model defaulting to majority classes (Safe, Violent Crimes) given the imbalance.

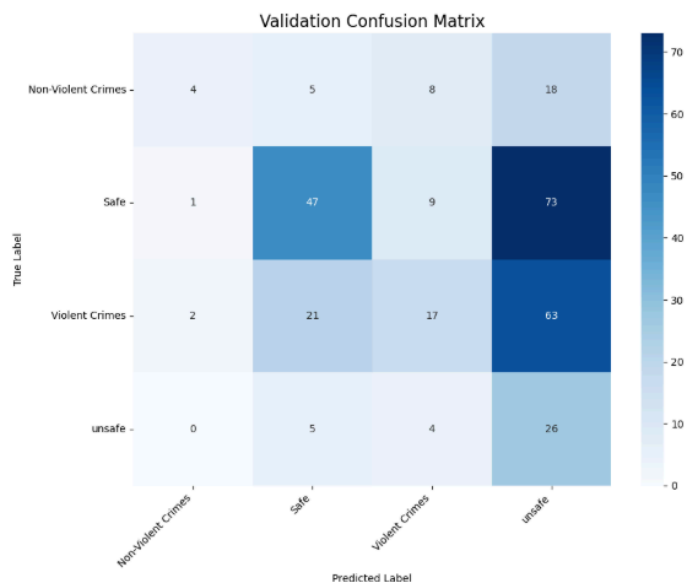
Results

Model performance was measured with accuracy and F1-score on train, validation, and test.

LSTM

Confusion Matrix – Train





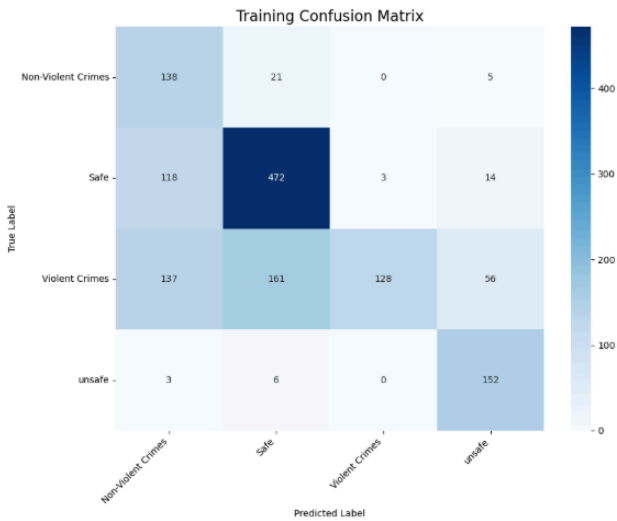
Confusion Matrix – Validation

Confusion Matrix – Test

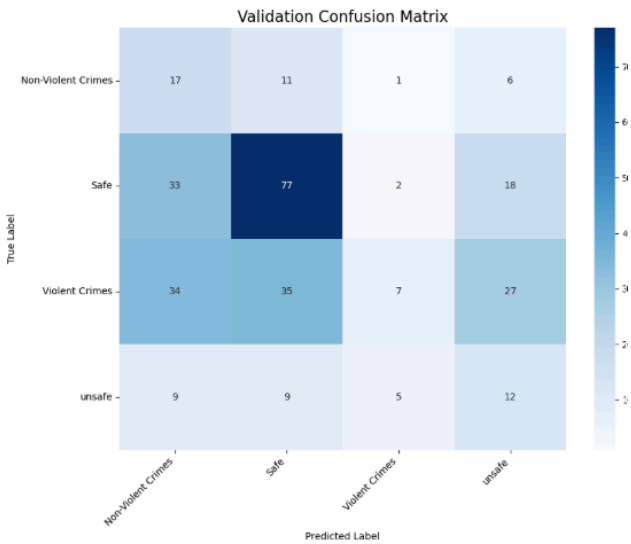


RNN

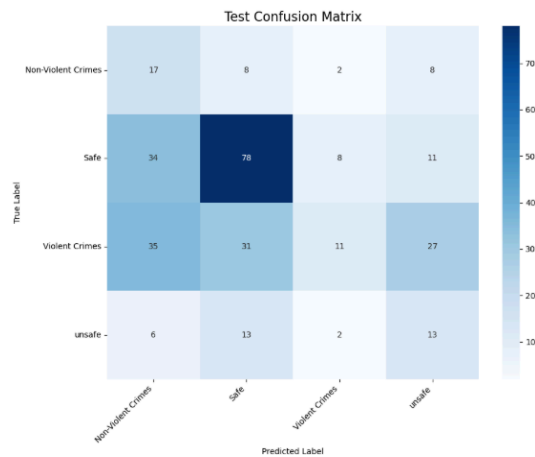
Confusion Matrix – Train



Confusion Matrix – Validation



Confusion Matrix – Test



Limitations & Next Steps

- **Suicide & Self-Harm** has only 3 unique examples after dedup — far too few to train or evaluate reliably. Sourcing more real examples for this class (rather than merging it away) would materially help, since it needs a distinct response path from generic "unsafe" content.
- **Dataset size.** After deduplication the usable dataset is roughly 2,000 unique texts, small for an LSTM trained from scratch; a pretrained embedding (e.g. GloVe) or a pretrained transformer encoder would likely outperform a from-scratch LSTM at this scale.
- **Class imbalance** remains even after merging (Safe and Violent Crimes dominate); class weights help but don't fully offset it — oversampling minority classes or collecting more data for Non-Violent Crimes / unsafe would help further.
- **Next step:** review the confusion matrices above once added — they'll show whether errors are concentrated in specific class pairs (e.g. Non-Violent Crimes vs. unsafe, which overlap conceptually) rather than spread evenly.